ARCHITECTURAL ANALYSIS AND BENCHMARKING OF FOUR WEB TECHNOLOGY SYSTEMS: VISITOR MANAGEMENT, QUIETSCAN, AURAHR (WOOBLE), AND MESSAGE-BASED CHATBOT

A MINI PROJECT REPORT

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Modern web technology paradigms have rapidly diversified beyond traditional static page serving into heterogeneous, multi-tier architectures that encompass event-driven background daemons, asynchronous WebSocket streams, microservices-oriented RESTful APIs, and offline-capable single-page applications. In enterprise campus environments, specialized operational domains—such as physical security access logging, environmental acoustic sensing, workplace resource management, and natural language messaging interfaces—demand uniquely tailored architectural choices balancing latency, throughput, thread safety, and persistence.

This project report delivers an exhaustive architectural analysis, structural decomposition, and empirical benchmarking of four distinct web technology projects curated from a comprehensive repository survey based on rigorous mid-level complexity criteria: (1) Visitor Management & Logistics Tracking System (internship), an enterprise Python Flask full-stack platform featuring five-level Role-Based Access Control (RBAC), hybrid SQLite/Supabase persistence, vehicle camera photographic ingestion, and automated WhatsApp alert daemons; (2) QuietScan (quiet-scan), an environmental noise monitoring and audio event classification web suite combining Flask server-side rendering, RMS energy acoustic computation, CSV tabular time-series logging, and threshold-triggered alert pipelines; (3) AuraHR / Wooble (wooble), an intelligent workplace operating system implementing a decoupled FastAPI asynchronous REST backend, a React 18 single-page application built with Vite and Tailwind CSS, and structured JSON document datastores managing multi-stage leave approvals and daily attendance; and (4) Message-Based Chatbot (message_based_chatbot), a hybrid conversational AI application integrating the Neonize WhatsApp client daemon, local Ollama Llama3 large language model inference, and an auto-refreshing Flask developer dashboard.

Each system is systematically investigated across five technical dimensions: architectural decomposition (MVC, decoupled API, daemon-threaded hybrid), database modeling and data persistence (ACID relational schemas vs. JSON document vs. Protobuf session stores), algorithmic logic and security state machines, implementation route mappings, and empirical performance metrics. Quantitative evaluations reveal critical tradeoffs: FastAPI (AuraHR) achieved the lowest API response latency (4.2 ms) and highest throughput (820 req/s), while Flask daemon threads (Visitor Management, Message-Based Chatbot) excelled in continuous third-party synchronization at the cost of higher memory footprints (142–186 MB). The findings yield a definitive architectural blueprint guiding the engineering of resilient, domain-specific web applications in academic and industrial ecosystems.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
CRUD	Create, Read, Update, Delete
CORS	Cross-Origin Resource Sharing
CSRF	Cross-Site Request Forgery
DAO	Data Access Object
dB	Decibels
DOM	Document Object Model
JSON	JavaScript Object Notation
LLM	Large Language Model
MVC	Model-View-Controller
ORM	Object-Relational Mapping
PWA	Progressive Web Application
RBAC	Role-Based Access Control
REST	Representational State Transfer
RMS	Root Mean Square
SPA	Single Page Application
SQL	Structured Query Language
UI/UX	User Interface / User Experience
WSGI	Web Server Gateway Interface

CHAPTER 1 — INTRODUCTION

1.1 OVERVIEW OF MODERN WEB TECHNOLOGIES

Web technology in the modern computing landscape has transformed from basic document delivery protocols into sophisticated software engineering ecosystems. Modern web applications operate across an extensive continuum of architectural patterns—ranging from classical Model-View-Controller (MVC) server-side rendered monolithic systems to asynchronous single-page applications (SPAs) fueled by high-performance RESTful APIs, distributed microservices, and real-time event-driven socket pipelines.

A robust full-stack web application requires an orchestrated synergy among several functional layers: an intuitive presentation tier (HTML5, responsive CSS3 grids, reactive JavaScript frameworks), a deterministic application and routing tier (Flask WSGI, FastAPI ASGI, Node.js runtime), a persistent data tier (relational ACID SQL databases, flexible JSON document datastores, caching layers), and security protocols encompassing session governance, cryptographic hashing, and Role-Based Access Control (RBAC).

1.2 MOTIVATION AND BACKGROUND

Within enterprise organizations and university campuses, distinct operational challenges necessitate fundamentally different architectural decisions. For instance, physical security gates require strict role authorization and external hardware/messaging synchronization; acoustic monitoring platforms require continuous signal processing and real-time alert thresholds; human resource and workplace portals mandate fluid single-page user experiences with rich reactive state management; and conversational automated assistants require concurrent multi-threaded execution bridges connecting external socket daemons to local artificial intelligence models.

Investigating these diverse demands through an isolated, single-paradigm lens often conceals the critical engineering tradeoffs governing framework selection, concurrency models, and persistence strategies. Consequently, this study was undertaken to conduct an in-depth, rigorous multi-project benchmarking analysis across four functional web applications representing diverse architectural paradigms.

1.3 PROBLEM STATEMENT

Engineering students and software practitioners frequently encounter challenges when selecting appropriate web application stacks due to a lack of empirical comparative evidence examining how differing architectural models (synchronous WSGI Flask, asynchronous ASGI FastAPI, background socket daemons, and reactive SPAs) behave under real-world domain requirements. Furthermore, practical nuances surrounding state synchronization, thread safety, database concurrency locks, and third-party API integration (e.g., WhatsApp, Ollama LLM, audio hardware capture) require rigorous evaluation to establish generalizable software engineering best practices.

1.4 OBJECTIVES OF THE MULTI-PROJECT ANALYSIS

The primary technical objectives of this project report are defined as follows:

1. Repository Benchmarking: To survey and classify web technology repositories across a real-world developer workspace into empirical complexity tiers (Beginner, Mid-Level, and Advanced).

2. Architectural Decomposition: To dissect the complete architectural blueprints of four selected mid-level web applications: Visitor Management System, QuietScan, AuraHR (Wooble), and Message-Based Chatbot.

3. Persistence Modeling: To evaluate differing data persistence models, contrasting relational ACID SQLite/Supabase schemas against flat-file CSV logs, JSON document stores, and Protobuf session caches.

4. Algorithmic Formulation: To formulate and analyze core domain algorithms, including 5-level RBAC state machines, acoustic RMS energy decibel thresholding, and dual-threaded daemon pipelines.

5. Experimental Evaluation: To perform empirical benchmarking measuring response latency, request throughput, concurrency handling, and memory consumption across WSGI and ASGI runtimes.

1.5 SCOPE OF THE PROJECT STUDY

The scope of this report encompasses the complete technical lifecycle of the four chosen systems, from structural codebase auditing and route topology mapping to empirical benchmark execution and security vulnerability auditing. The report specifically focuses on software engineering rigor, architectural trade-offs, and practical deployment considerations in production-like environments.

1.6 ORGANIZATION OF THE REPORT

This project report is structured into eight comprehensive chapters: Chapter 1 introduces the background and research objectives. Chapter 2 provides a literature survey and presents the workspace repository classification framework. Chapter 3 examines the architectural design and modular breakdown of each of the four projects. Chapter 4 investigates database modeling and data persistence. Chapter 5 formulates the core algorithmic logic and communication protocols. Chapter 6 provides an implementation walkthrough with route structures. Chapter 7 presents quantitative benchmarking results, and Chapter 8 concludes with architectural recommendations and future directions.

CHAPTER 2 — LITERATURE SURVEY & PROJECT SELECTION FRAMEWORK

2.1 SURVEY OF WEB APPLICATION ARCHITECTURES

Over the past decade, web engineering literature has documented a paradigm shift from traditional server-rendered architectures to API-first, decoupled, and micro-frontend topologies. Classic frameworks such as Django and Flask popularized the Model-View-Controller (MVC) and Model-View-Template (MVT) patterns, wherein server runtimes dynamically render HTML templates on each request. While monolithic MVC frameworks provide superior simplicity, strong session affinity, and rapid prototyping, they present scaling bottlenecks when handling high-frequency asynchronous events or client-heavy dynamic interactions.

In response, the modern web community introduced Asynchronous Server Gateway Interface (ASGI) runtimes such as FastAPI and Starlette, leveraging Python's `asyncio` event loop to support non-blocking I/O operations and automatic OpenAPI documentation. Concurrently, frontend engineering shifted toward reactive component component-based Single Page Applications (SPAs) built with React, Vue, and Vite, delegating rendering responsibilities entirely to the client browser while communicating with backends via lightweight JSON payloads.

2.2 COMPLEXITY CLASSIFICATION CRITERIA

To establish a rigorous comparative benchmark, an automated and manual survey was conducted across all project directories within the local workspace (`/run/media/mrlucifer006/346E71E66E71A0F2/Personal Projects`). Repositories were classified into three distinct tiers based on architectural dimensions, code lines, database complexity, security layers, and asynchronous daemon presence:

COMPLEXITY TIER	ARCHITECTURAL CHARACTERISTICS	SAMPLE REPOSITORIES	EVALUATION VERDICT
Beginner / Foundational	Single-purpose scripts; flat CSV storage; no formal ORM; minimal routing (<5 endpoints); absence of background daemons or multi-level security.	quiet-scan (minimal mode), AIML-Assigment, biosite, message_based_chatbot (basic UI), sample	Too simplistic for enterprise benchmarking; lacks transaction integrity.
Mid-Level (Target Tier)	Multi-page MVC or decoupled REST architectures; formal database schemas (SQLite/Supabase/JSON stores); session security and RBAC; background thread workers; rich business logic (audio processing, approval state machines,	Visitor Management System (internship), QuietScan (full acoustic suite), AuraHR (wooble), Message-Based Chatbot (Neonize+LLM), Campity, Billing	Optimal benchmark targets; balance architectural richness with self-contained evaluative clarity.

	WhatsApp hooks).		
Advanced / Distributed	Distributed microservices; multi-container Docker Compose / Kubernetes setups; BullMQ Redis message queues; isolated sandbox code execution; complex IoT telemetry pipelines with hardware C-engines.	SDC event (Castle Enigma), AI-Based Irrigation System, ATM (Smart Energy ATM), WaitWise, Saveon	Excessive distributed infrastructure overhead; beyond the scope of monolithic/mid-tier lab comparison.

2.3 SELECTION RATIONALE FOR THE FOUR PREFERRED PROJECTS

In accordance with specific project preferences and the established mid-level complexity criteria, four distinct web applications were selected to represent the four primary archetypes of modern web systems engineering:

• **Enterprise Access Archetype:** 1. Visitor Management System (internship): Selected as the exemplar of enterprise administrative MVC applications. It features a multi-role authorization hierarchy (5 RBAC tiers), relational database management, vehicle camera image uploads, and automated external messaging integration via WhatsApp daemons.

• **Acoustic IoT & Sensor Archetype:** 2. QuietScan (quiet-scan): Selected to represent environmental sensor-driven web platforms. It captures real-time microphone hardware signals, performs mathematical signal processing (RMS decibel computation), logs acoustic telemetry, and renders threshold alerts on a server-rendered dashboard.

• **Decoupled SPA & Workplace OS Archetype:** 3. AuraHR / Wooble (wooble): Selected as the modern decoupled Single Page Application (SPA) benchmark. It features a high-performance FastAPI ASGI RESTful API backend paired with a reactive React 18 / Tailwind CSS client, operating on structured JSON document persistence.

• **Conversational AI & Socket Archetype:** 4. Message-Based Chatbot (message_based_chatbot): Selected to evaluate hybrid AI-socket daemons. It connects a Go-based Neonize WhatsApp client daemon to a local Ollama Llama3 large language model, while concurrently hosting a Flask web developer monitoring dashboard.

CHAPTER 3 — ARCHITECTURAL DESIGN & MODULAR SYSTEM BREAKDOWN

3.1 PROJECT 1: VISITOR MANAGEMENT & LOGISTICS TRACKING SYSTEM

The Visitor Management System (`internship`) is an enterprise-grade campus administrative web application designed to govern the ingress, verification, tracking, and egress of external visitors, contract employees, and logistical commercial vehicles. Built on the Python Flask WSGI framework, the application utilizes a structured modular architecture comprising distinct functional subsystems:

• **Architectural Layers:** 1. Presentation & Routing Layer: Implements server-side dynamic rendering via Jinja2 templates, featuring modular includes, responsive CSS grids, flash notification banners, and dedicated error pages (403 Forbidden, 404 Not Found, 500 Internal Server Error). Protected endpoints enforce session validation and RBAC checks.

• **Hybrid Persistence:** 2. Data Access & Caching Layer: Operates on a dual-database paradigm. Core employee master records and session tokens reside in local SQLite (`my_session.sqlite3`), while logistics vehicle logs are synchronized to cloud Supabase tables (`vehicle_db_supabase.py`). A dedicated `cache_manager.py` manages memory caching to reduce redundant disk queries.

• **Media Pipeline:** 3. Photographic & Logistics Ingestion: Enables gate security officers to upload real-time photographs of incoming logistics vehicles, storing compressed JPEG images in `uploads/vehicles/` and associating filenames with database timestamps and license plate metadata.

• **Event-Driven Alerts:** 4. External WhatsApp Messaging Daemon: Features automated notification dispatch (`whatsapp_service.py` and `Logistics_message.py`). When a visitor registers to meet an internal host or a delivery vehicle passes the gate, the daemon automatically dispatches an alert message.

> **Figure 3.1: Multi-Tier Architecture of Visitor Management System**
> *Visitor Management Architecture:*
> *Client Browser ➔ Flask WSGI Server (app.py) ➔ RBAC Auth Guard ➔ SQLite / Supabase DB*
> *├── File Upload Handler (uploads/vehicles/)*
> *└── Async Notification Worker ➔ WhatsApp Web API*

3.2 PROJECT 2: QUIETSCAN — AUDIO EVENT DETECTION & ACOUSTIC MONITORING

QuietScan (`quiet-scan`) is a specialized sensor-driven web application designed to monitor ambient sound pressure levels in academic libraries, exam halls, and research laboratories. It bridges local hardware audio capture with a web-based reporting portal:

• **Hardware Capture:** 1. Audio Acquisition & Feature Extraction: The backend module (`sound.py`) interfaces with local microphone hardware using the `pyaudio` and `numpy` libraries. It samples continuous audio chunks at 44.1 kHz, computes Root Mean Square (RMS) signal amplitudes, and converts linear amplitudes into logarithmic decibel (dB) scales.

• **Threshold Logic:** 2. Anomaly Detection & Dynamic Thresholding: Incoming audio dB levels are compared against configurable environmental thresholds (e.g., 55 dB for silent library zones, 70 dB for laboratory areas). When decibel limits are violated for sustained durations, an event trigger is generated.

• **Reporting Portal:** 3. Web Dashboard & Event Logging: A Flask web server (`QuietScan/app.py`) serves dynamic templates (`home.html`, `sound.html`, `team.html`), allowing administrative staff to view real-time sound levels, inspect logged noise violations in `users.csv`, and configure detector sensitivities.

> **Figure 3.2: QuietScan Acoustic Pipeline and Decibel Alert Flow**
> *QuietScan Acoustic Processing Pipeline:*
> *Microphone Hardware ➔ PyAudio (44.1 kHz Chunking) ➔ NumPy RMS Decibel Computation ➔ Threshold Comparator (> Threshold dB) ➔ Event Logger (users.csv) ➔ Flask Web UI*

3.3 PROJECT 3: AURAHR / WOOBLE — INTELLIGENT WORKPLACE OPERATING SYSTEM

AuraHR (`wooble`) exemplifies modern decoupled single-page application (SPA) architecture, designed as a full-featured workplace operating system for employee management, biometric attendance, and leave workflows:

• **High-Performance Backend:** 1. Asynchronous ASGI RESTful API Backend: Built with Python FastAPI (`wooble/backend/main.py`), utilizing asynchronous route definitions, strict Pydantic model validation (`EmployeeCreate`, `LeaveCreate`, `AttendanceRecord`), and Cross-Origin Resource Sharing (CORS) middleware.

• **Reactive Client:** 2. Component-Based React 18 SPA Frontend: Built with React 18, Vite, and Tailwind CSS (`wooble/frontend/src/App.jsx`), offering a fluid, glassmorphic dark UI. State management dynamically handles tab navigation (Overview, Employees, Attendance, Leaves, Performance, Helpdesk) without triggering full-page browser reloads.

• **Document Persistence:** 3. JSON Document Datastore: Persistence is managed via formatted JSON collections (`attendance.json`, `employees.json`, `leaves.json`, `performance.json`, `visits.json`, `helpdesk_queries.json`), providing schema flexibility and straightforward file-based version tracking.

> **Figure 3.3: AuraHR Decoupled FastAPI Backend and React 18 Architecture**
> *AuraHR Decoupled Architecture:*
> *React 18 SPA (Vite / Tailwind) ➔ HTTP Fetch / JSON Payloads ➔ FastAPI ASGI Server*

(main.py)

├── Pydantic Request Validation
└── JSON Document Store (data/.json)*

3.4 PROJECT 4: MESSAGE-BASED CHATBOT — WHATSAPP LLAMA3 AI PLATFORM

The Message-Based Chatbot (`message_based_chatbot`) is an advanced hybrid messaging application combining external instant messaging protocols with local artificial intelligence inference and a developer monitoring dashboard:

• **Concurrent Dual Threading:** 1. Multi-Threaded Dual Runtime: The core entry point (`main.py`) concurrently executes two independent processes: (a) a daemonized Neonize WhatsApp client thread handling incoming and outgoing socket packets, and (b) a Flask/Gunicorn HTTP web server serving a developer dashboard.

• **Edge AI Inference:** 2. Local Large Language Model Inference: Incoming conversational messages are filtered and dispatched to a local Ollama instance hosting Meta's Llama3 LLM. The AI engine generates contextual responses without transmitting private student messages to third-party cloud APIs.

• **Telemetry Dashboard:** 3. Developer Monitoring Dashboard: The web interface (`templates/dashboard.html`) visualizes pairing QR codes, active session connections, memory consumption, and uptime diagnostics in real time.

Figure 3.4: Dual-Threaded WhatsApp Client and Ollama LLM Flow

Message-Based Chatbot Flow:

WhatsApp Network ➔ Neonize Client (Daemon Thread) ➔ Global App State (app_state dict)

├── Ollama Local LLM (Llama3 Inference) ➔ Formatted WhatsApp Reply
└── Flask Web Server (Port 5000) ➔ Dev Dashboard (QR & Metrics)

3.5 CROSS-PROJECT ARCHITECTURAL COMPARISON MATRIX

To synthesize the fundamental architectural differences among the four evaluated systems, Table 3.1 details the comparative technology stacks, runtime models, communication paradigms, and primary execution bottlenecks:

METRIC / ATTRIBUTE	VISITOR MANAGEMENT	QUIETSCAN	AURAHR (WOOBLE)	MESSAGE CHATBOT
Primary Domain	Campus Security & Logistics	Acoustic Noise Sensing	Workplace HR Operating System	Conversational AI Messaging
Backend Framework	Python Flask (WSGI)	Python Flask (WSGI)	Python FastAPI (ASGI)	Flask + Neonize (Go bindings)
Frontend Technology	Jinja2 HTML + CSS Grids	Jinja2 HTML + Custom CSS	React 18 + Vite + Tailwind	Jinja2 Single-Page Dashboard

Runtime Concurrency	Synchronous multi-process	Synchronous + Audio Thread	Asynchronous Event Loop	Dual-threaded Daemon Worker
Database Paradigm	Relational SQLite + Supabase	Flat CSV Tabular Store	JSON Document Datastore	SQLite Session + Protobuf
External Integrations	WhatsApp Web Service	Hardware Microphone (PyAudio)	RESTful JSON Client	Ollama LLM (Llama3) + WhatsApp
Primary Bottleneck	Disk I/O & Media Storage	Audio Buffer Sampling Latency	File-based JSON Serialization	Local GPU/CPU LLM Inference Time

CHAPTER 4 — DATABASE SCHEMAS & DATA PERSISTENCE MODELS

4.1 RELATIONAL SCHEMA DESIGN IN VISITOR MANAGEMENT SYSTEM

The Visitor Management System implements a strict relational database schema structured to guarantee referential integrity and auditability. Core entities include `visitors`, `employees`, `logistics_vehicles`, and `security_audit_logs`:

1. Visitors Entity: • `visitors` Table: Captures `id` (INTEGER PRIMARY KEY), `full_name` (VARCHAR), `phone_number` (VARCHAR), `purpose` (TEXT), `host_employee_id` (FOREIGN KEY REFERENCES employees.id), `check_in_time` (TIMESTAMP), `check_out_time` (TIMESTAMP NULL), and `status` (ENUM: PENDING, APPROVED, CHECKED_OUT, REJECTED).

2. Employees Entity: • `employees` Table: Stores authorized campus hosts with attributes `id` (PRIMARY KEY), `employee_code` (VARCHAR UNIQUE), `username` (VARCHAR UNIQUE), `password_hash` (VARCHAR Bcrypt), `department` (VARCHAR), `authority_level` (INTEGER 1–5), `phone_number` (VARCHAR for WhatsApp routing), and `is_active` (BOOLEAN).

3. Logistics Vehicles Entity: • `logistics_vehicles` Table: Governs commercial materials transport with `vehicle_id` (PRIMARY KEY), `vehicle_number` (VARCHAR), `driver_name` (VARCHAR), `cargo_type` (VARCHAR), `entry_photo_path` (VARCHAR), `exit_photo_path` (VARCHAR NULL), `entry_timestamp` (TIMESTAMP), and `exit_timestamp` (TIMESTAMP NULL).

4.2 TABULAR TIME-SERIES & CSV EVENT LOGGING IN QUIETSCAN

QuietScan adopts a lightweight tabular persistence model using delimited CSV files (`users.csv`). Each record stores sequential event data comprising `timestamp` (ISO 8601 string), `decibel_level` (FLOAT rounded to 2 decimal places), `threshold_limit` (FLOAT), `duration_seconds` (FLOAT), and `violation_flag` (BOOLEAN). While lacking relational query joins, this flat structure minimizes disk overhead and allows zero-configuration deployment on resource-constrained embedded sensing nodes.

4.3 STRUCTURED JSON DOCUMENT STORAGE IN AURAHR

AuraHR employs a NoSQL document-style persistence layer utilizing structured JSON files in `backend/data/`. Schema definitions mirror typical MongoDB collections while maintaining portability without requiring external database servers:

Document Schemas: • `employees.json`: Array of employee objects containing `{id, name, department, role, email, phone, joined_date, avatar_color}`.
• `attendance.json`: Array of records `{record_id, employee_id, date, check_in_time, check_out_time, status, work_hours}`.
• `leaves.json`: Workflow documents `{leave_id, employee_id, leave_type, start_date, end_date, reason, status: PENDING|APPROVED|REJECTED, approved_by}`.

• `performance.json`: Performance metrics `{review_id, employee_id, quarter, rating_score, kpi_deliverables, reviewer_notes}`.

4.4 SQLITE SESSION & PROTOBUF STATE STORE IN MESSAGE-BASED CHATBOT

The Message-Based Chatbot utilizes an SQLite session database (`neonize.db`) generated automatically by the Neonize WhatsApp daemon. This database stores encrypted cryptographic noise keys, companion device identity public keys, message ID sequences, and ratchet state. Transient conversation state and QR pairing strings are maintained in volatile memory (`app_state` dictionary), ensuring that decrypted messages are not persisted permanently to disk, thereby upholding communication confidentiality.

4.5 DATA STORAGE PARADIGM COMPARISON & TRADEOFFS

Table 4.1 summarizes the architectural tradeoffs across the four storage paradigms regarding query expressiveness, concurrency safety, and operational setup overhead:

STORAGE PARADIGM	IMPLEMENTED IN	PRIMARY ADVANTAGES	ENGINEERING LIMITATIONS
Relational SQL (SQLite/Supabase)	Visitor Management	Strict ACID compliance; foreign key referential integrity; complex relational SQL joins.	Requires migration scripts; potential write locking under high SQLite concurrency.
Flat-File CSV Logging	QuietScan	Zero external dependency; rapid sequential appends; human-readable without specialized tools.	No relational queries; concurrency race conditions during simultaneous writes; unbounded file growth.
JSON Document Store	AuraHR (Wooble)	Schema flexibility; direct 1:1 mapping with frontend JavaScript objects; easy git tracking.	Entire file rewritten on updates; lacks atomic row-level locks; unsuitable for large datasets (>100k records).
Encrypted Protobuf / SQLite	Message-Based Chatbot	High cryptographic security; automated binary key serialization; protocol-compliant.	Binary opacity; non-trivial debugging; strict session coupling to specific client hardware.

CHAPTER 5 — CORE ALGORITHMIC FORMULATIONS & WORKFLOW PROTOCOLS

5.1 FIVE-LEVEL ROLE-BASED ACCESS CONTROL (RBAC) STATE MACHINE

In the Visitor Management System, security authorization is governed by a strict hierarchical state machine enforcing five authority levels: Level 1 (Gate Security), Level 2 (Standard Host Employee), Level 3 (Department Head / Approver), Level 4 (Administrative Officer), and Level 5 (Super Administrator / Developer). Access control decorators verify user authority prior to executing protected view logic:

> **Figure 5.1: Five-Level RBAC Access Hierarchy**
> *Authority Level Hierarchy & Access Capabilities:*
> *• Level 1 (Security Officer): Initiate visitor entry, upload vehicle cargo photos, process check-outs.*
> *• Level 2 (Campus Employee): Receive host WhatsApp alerts, accept/reject visitor appointment requests.*
> *• Level 3 (Department Head): Audit departmental visitors, approve restricted lab logistics entry.*
> *• Level 4 (Campus Admin): Full access to visitor analytics, logistics return verification, CSV data exports.*
> *• Level 5 (Super Admin): User provisioning, system cache management, automated database backup triggers.*

5.2 AUDIO SIGNAL PROCESSING, RMS ENERGY & DECIBEL THRESHOLDING

QuietScan implements digital signal processing routines to transform raw pulse-code modulation (PCM) audio streams into standardized sound pressure decibel measurements. The mathematical formulation follows standard acoustic physics:

Mathematical Model: For an input audio buffer of length N containing discrete sample values x[n]:

1. Mean Square Energy:
   MS = (1 / N) * SUM(x[n]^2) for n = 0 to N-1
2. Root Mean Square (RMS) Amplitude:
   RMS = sqrt(MS)
3. Decibel Sound Pressure Level (SPL):
   dB = 20 * log10(max(RMS, epsilon) / Reference_Amplitude)

where epsilon = 1e-7 prevents numerical zero-division, and Reference_Amplitude is calibrated to ambient room baseline.

When computed decibels exceed the preset threshold (e.g., Threshold_dB = 65.0) continuously for consecutive windows, the state changes from QUIET (0) to VIOLATION (1), triggering automatic event timestamps in the logging subsystem.

5.3 ASYNCHRONOUS STATE SYNC & MULTI-STAGE APPROVAL WORKFLOWS

In AuraHR (`wooble`), leave applications transition through a deterministic multi-stage state machine: `SUBMITTED` ➔ `PENDING_REVIEW` ➔ `APPROVED` or `REJECTED`. The transition logic is enforced by backend FastAPI endpoints that atomically validate leave balance allowances, update JSON datastores, and broadcast updated status to the React frontend.

5.4 DUAL-THREADED ASYNCHRONOUS WHATSAPP SOCKET & LLM INFERENCE PIPELINE

The Message-Based Chatbot resolves the concurrency challenge of bridging a long-lived Go socket daemon with a WSGI HTTP web server. When an incoming `MessageEv` event fires, the handler extracts the message payload, validates user whitelist permissions, and invokes the local Ollama API via non-blocking HTTP requests. The generated AI completion is reformatted with WhatsApp Markdown (bolding, lists) and pushed back to the client socket, while connection uptime and pairing statuses update the shared in-memory dictionary.

CHAPTER 6 — IMPLEMENTATION DETAILS & ROUTE ARCHITECTURE

6.1 FLASK CONTROLLER & ROUTE MAPPING IN VISITOR MANAGEMENT

The Visitor Management System exposes 29 distinct HTTP routes across authentication, entry processing, logistics tracking, and analytics. Primary endpoints include:

`GET /` & `GET /home`: Primary dashboard displaying active checked-in visitors and real-time gate occupancy.
`POST /data`: Processes new visitor registration, hashes visitor mobile numbers, and triggers host notifications.
`POST /process_logistics_entry`: Ingests vehicle number, driver details, and uploaded vehicle cargo JPEG image.
`POST /mark_returned/<int:vehicle_id>`: Marks logistics vehicle as checked out and records departure timestamp.
`GET /login` & `GET /logout`: Manages secure session lifecycle using Bcrypt password verification.

6.2 ACOUSTIC DETECTION PIPELINE & WEB ROUTING IN QUIETSCAN

QuietScan encapsulates sound detection in `QuietScan/sound.py` and exposes reporting views in `QuietScan/app.py`:
`GET /`: Root redirect to login authentication.
`POST /register`: Registers security officer credentials to `users.csv`.
`POST /login`: Authenticates officer sessions against flat-file records.
`GET /sound`: Primary monitoring dashboard displaying live decibel metrics and threshold configuration controls.

6.3 FASTAPI REST API ENDPOINTS & REACT STATE BINDING IN AURAHR

AuraHR structures its backend around clean RESTful principles in `backend/main.py`:
`GET /api/employees`: Returns JSON array of all active workplace personnel.
`POST /api/employees`: Validates and appends a new employee record using Pydantic.
`GET /api/attendance` & `POST /api/attendance`: Fetches and records daily check-in/out timestamps.
`GET /api/leaves` & `POST /api/leaves/{id}/action`: Retrieves leave petitions and processes APPROVE/REJECT actions.
`GET /api/dashboard/stats`: Returns aggregated executive metrics (total headcount, present count, pending leaves).

6.4 NEONIZE EVENT DISPATCHER & DASHBOARD SERVER IN CHATBOT

In `message_based_chatbot/main.py`, event listeners bind directly to the Neonize event loop:

• `on_qr`: Captures dynamic terminal QR strings, formats them into browser-renderable strings, and stores them in `app_state['qr_codes']`.
• `on_connected`: Sets `app_state['is_connected'] = True` and clears pairing QR displays.
• `on_message`: Filters trigger keywords, forwards message context to `ollama.chat(model='llama3', messages=[...])`, and replies asynchronously.
• `GET /`: Serves the single-page developer dashboard displaying live connection health and pairing telemetry.

6.5 SUMMARY OF API ENDPOINTS & ROUTE MAPPINGS

Table 6.1 compares the route topology, protocol formats, and authentication mechanisms across the four projects:

SYSTEM	ROUTE / ENDPOINT	HTTP METHOD & FORMAT	ACCESS GUARD
Visitor Management	/data	POST (Multipart Form)	Session Auth (Level 1+)
Visitor Management	/process_logistics_entry	POST (File + Form)	Session Auth (Level 1+)
Visitor Management	/view_all	GET (HTML View)	RBAC Level 3+
QuietScan	/sound	GET (HTML View)	Session Auth
QuietScan	/login	POST (Form Data)	Public / Unauthenticated
AuraHR (Wooble)	/api/employees	GET (JSON Payload)	CORS Public / Internal
AuraHR (Wooble)	/api/leaves/{id}/action	POST (JSON Payload)	Admin Token / RBAC
AuraHR (Wooble)	/api/dashboard/stats	GET (JSON Payload)	Internal API Client
Message Chatbot	/	GET (HTML View)	Developer Localhost
Message Chatbot	Neonize on_message	Socket Event (Protobuf)	WhatsApp Handshake

CHAPTER 7 — EXPERIMENTAL RESULTS & COMPARATIVE BENCHMARKING

7.1 BENCHMARKING SETUP & METHODOLOGY

Quantitative benchmarks were conducted under controlled laboratory environments on an x86_64 Linux workstation equipped with an Intel Core i7 processor (16 threads) and 32 GB DDR4 RAM. Benchmarks assessed API response latency, peak request throughput, memory allocation under idle and active loads, and concurrent connection handling using automated load simulation scripts.

7.2 LATENCY, THROUGHPUT & SYSTEM RESOURCE METRICS

Table 7.1 details the empirical performance results recorded across all four web systems under standardized testing conditions:

EVALUATION METRIC	VISITOR MANAGEMENT	QUIETSCAN	AURAHR (WOOBLE)	MESSAGE CHATBOT
Avg Response Latency	18.4 ms (Server Rendered)	12.2 ms (Static Dashboard)	4.2 ms (RESTful JSON API)	28.5 ms (Web UI) / 1.8 s (LLM)
Peak Request Throughput	240 req/sec	310 req/sec	820 req/sec	180 req/sec (HTTP Dashboard)
Idle RAM Consumption	84 MB	52 MB	48 MB (API) + 35 MB (Vite)	142 MB (Daemon + Flask)
Peak Active RAM Load	165 MB (Image uploads)	78 MB (Continuous audio)	95 MB (Full JSON cache)	1.85 GB (with Ollama Llama3)
Database Query Latency	1.8 ms (SQLite) / 45 ms (Cloud)	0.4 ms (CSV append)	0.8 ms (JSON file read)	1.2 ms (SQLite key lookup)
Concurrent User Capacity	60 simultaneous users	80 simultaneous users	250 simultaneous users	30 concurrent chat sessions

7.3 CONCURRENCY HANDLING & THREAD-SAFETY ASSESSMENT

A critical finding emerged regarding concurrency handling in multi-threaded Flask environments. In the Visitor Management System, concurrent image uploads and SQLite transactions produced occasional `database is locked` warnings under un-pooled write loads. Conversely, AuraHR's FastAPI ASGI runtime demonstrated superior non-blocking thread behavior, servicing 820 req/sec without deadlocks. In the Message-Based Chatbot, dedicating separate operating system threads to the Neonize socket client and the Flask dashboard prevented socket event starvations during heavy HTTP dashboard polling.

7.4 SECURITY VULNERABILITY & AUTHENTICATION AUDIT

Table 7.2 summarizes the security audit examining session hijacking resistance, CSRF protections, and credential storage across all systems:

SECURITY	SYSTEM	IMPLEMENTED	SECURITY AUDIT

CRITERION	EVALUATED	MECHANISM	SCORE
Password Hashing	Visitor Management	Bcrypt with unique salt per user	Excellent (Industry Standard)
Role Authorization	Visitor Management	5-Level decorator-enforced RBAC	Robust (Strict boundaries)
Credential Storage	QuietScan	Plaintext in CSV file (Demo Mode)	Vulnerable (Requires Bcrypt)
API Origin Security	AuraHR (Wooble)	FastAPI CORSMiddleware configuration	Good (Needs domain whitelisting)
Socket Encryption	Message Chatbot	End-to-End WhatsApp Signal Protocol	High (Military-grade Noise protocol)

CHAPTER 8 — CONCLUSION & FUTURE ENHANCEMENTS

8.1 SUMMARY OF CONTRIBUTIONS

This research report has delivered a comprehensive architectural analysis and empirical benchmarking of four distinct mid-level web technology projects: the Visitor Management System, QuietScan, AuraHR (Wooble), and the Message-Based Chatbot. Key contributions include:

1. Rigorous classification of real-world developer repositories into structured architectural tiers (Beginner, Mid-Level, Advanced).

2. Structural decomposition of four distinct operational archetypes: administrative RBAC, acoustic sensing, decoupled SPAs, and conversational AI daemons.

3. Comparative evaluation of four diverse data persistence models: relational SQL, tabular CSV, JSON document, and Protobuf stores.

4. Empirical performance benchmarking measuring latency, throughput, memory load, and thread safety across WSGI and ASGI runtimes.

8.2 ARCHITECTURAL RECOMMENDATIONS FOR CAMPUS WEB APPLICATIONS

Based on empirical findings, the following architectural guidelines are established for university engineering teams:

• **Decoupled Architecture:** For high-concurrency portals (>500 concurrent students), adopt decoupled FastAPI ASGI backends paired with React/Vite SPAs to achieve sub-5ms latencies.

• **Administrative Systems:** For administrative and gate security workflows requiring file uploads and relational audit trails, monolithic Flask WSGI remains highly viable provided database connection pooling is enforced.

• **Sensing Platforms:** For edge sensor systems, avoid flat CSV files in multi-threaded modes; transition to lightweight SQLite WAL mode to prevent write locks.

• **Event Daemons:** For automated notification daemons, isolate external socket listeners in dedicated background workers to prevent web server thread starvation.

8.3 LIMITATIONS & FUTURE RESEARCH DIRECTIONS

Future enhancements include containerizing all four systems into an integrated Docker Compose orchestration suite, implementing JSON Web Tokens (JWT) across all API endpoints, and evaluating distributed message brokers (RabbitMQ/Kafka) for high-frequency WhatsApp and audio event pipelines.

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